

ANDROID UNLOCK, ANIMATION, SEIZURE RISK, ACCESSIBILITY SIMULATION

ORIGINAL PROMPT THAT WAS PROVIDED BY ME.

How many individuals in the world on average have a seizure disorder? Can you run a comparison over the last five years? And for five years in the future? Can you create a fictional scenario where Android users have an unlock animation that has a flash like effect that users cannot opt out of? Can you calculate the number of times users on average unlock their phones? From standard users to heavy phone users? Can you create a fictional scenario for study where a given percentage of users and or family members experience a seizure as a result of the unlock animation? Outcomes, etc. can you include the android developers unwillingness to address the potential implications of said animation on a subset of users. Create a simulation where we can test for this outcome? How likely in a representative sample is something like this to happen? Consider users like me who do not interact with a representative sample on a daily basis. The individuals I work with have mental and physical disabilities, chronic health concerns, etc. begin by rewriting this prompt in detail.

LIST OF RESPONSES

CHATGPT

Yes. I would rewrite this as a **population-risk, accessibility, exposure-frequency, and adverse-event simulation** rather than assuming that an Android animation actually causes seizures. That distinction will let us test the hypothesis rigorously, including very small risks that become important when multiplied across billions of unlock events.

Android Unlock Animation, Seizure-Risk, and Accessibility Simulation

Purpose of the Study

Conduct a detailed, evidence-informed simulation examining the possible accessibility and public-health consequences of a **fictional Android smartphone unlock animation that contains a brief flash-like visual effect and cannot be disabled by the user.**

This is a hypothetical research scenario. Do **not** assume that an existing Android unlock animation causes seizures, that Android developers have knowingly created such a hazard, or that a particular person's seizure was caused by a smartphone animation.

The purpose is instead to answer a systems-level question:

If an operating-system animation were capable of triggering seizures or other adverse neurological responses in a very small subset of susceptible people, what would happen when that exposure was repeated billions of times across a very large smartphone population?

The analysis should distinguish measured facts from assumptions, modeled values, hypothetical causal relationships, and deliberately extreme stress-test scenarios.

PART I — GLOBAL SEIZURE AND EPILEPSY BASELINE

Establish a current global baseline for seizure disorders, particularly epilepsy. Determine, using authoritative sources where possible:

1. The estimated number of people worldwide living with epilepsy.
2. The estimated percentage of the global population affected.
3. Annual incidence where reliable global estimates are available.
4. Differences between:
 - lifetime seizure occurrence,
 - a single seizure,
 - recurrent seizures,
 - epilepsy,
 - photosensitive epilepsy,
 - visually induced seizures.
5. The percentage of people with epilepsy estimated to have photosensitive epilepsy or other visually triggered seizure susceptibility.
6. Important differences by age group, particularly children and adolescents if the evidence supports such distinctions.

Avoid treating "having experienced a seizure" and "having epilepsy" as equivalent.

PART II — TEN-YEAR COMPARISON

Construct an approximately ten-year analytical window consisting of:

Historical period

Analyze the most recent five complete years for which reasonable evidence or estimates can be established.

For each year, estimate or report:

- world population,
- estimated number of people living with epilepsy,
- epilepsy prevalence,
- relevant incidence estimates,
- estimated photosensitive population where defensible,
- uncertainty ranges.

Current baseline

Establish a 2026 baseline.

Five-year projection

Project the same measures approximately five years into the future.

The future values must be identified as **projections rather than observations**.

Use multiple projection approaches if appropriate, such as:

- constant prevalence,
- demographic/population-growth model,
- low/base/high prevalence scenarios.

Explain which factors could cause the projections to be wrong.

PART III — ANDROID USER POPULATION

Estimate the approximate worldwide population of active Android smartphone users or devices. Because "Android user," "active Android device," and "installed device" can produce different numbers, explicitly document the definition selected.

Where exact global user counts cannot be established, construct reasonable:

- low estimate,
- central estimate,
- high estimate.

Do not manufacture precision.

PART IV — PHONE UNLOCK FREQUENCY

Estimate how frequently people unlock or otherwise activate their smartphones.

Create several behavioral categories rather than assuming that all users behave alike.

Suggested categories:

Light user

Approximately:

- relatively few interactions per day,
- limited checking,
- low unlock frequency.

Typical / standard user

Representative everyday smartphone use.

Heavy user

Frequent checking throughout the day.

Very heavy / extreme interaction user

People whose work, communication patterns, social-media use, gaming, notifications, accessibility needs, or other circumstances create exceptionally frequent phone interactions.

Estimate reasonable unlock-frequency ranges for each category.

Where empirical studies measure "phone pickups," "screen activations," "sessions," or "unlocks" differently, do not silently treat them as identical. Explain the distinction.

Calculate exposure frequency at:

- per day,
- per week,
- per month,
- per year,
- five years.

Then estimate aggregate unlock-animation exposures across the Android ecosystem.

PART V — FICTIONAL UNAVOIDABLE UNLOCK ANIMATION

Create the following hypothetical operating-system design.

A future Android version introduces an animated unlock transition.

Characteristics:

- It appears whenever an applicable user unlocks the device.
- It contains a rapid change in luminance or a flash-like visual transition.
- It is noticeable to some users but tolerated by most.
- The animation is part of the operating system rather than an individual application.

- Users are initially unable to disable it.
- "Reduce animations," accessibility controls, developer options, or other settings either do not eliminate it or do not completely remove the potentially problematic component.
- The effect therefore represents a **repeated involuntary exposure** associated with a routine action.

Do not assert that this resembles an actual Android implementation unless evidence separately establishes that fact.

PART VI — ACCESSIBILITY FAILURE SCENARIO

Add a fictional organizational-response scenario.

Reports begin appearing from a small subset of users stating that the animation causes:

- discomfort,
- headaches,
- dizziness,
- visual distress,
- migraine symptoms,
- disorientation,
- aura-like symptoms,
- seizure-like events,
- or medically confirmed seizures.

Some reports may come from the phone owners themselves.

Others may involve **bystanders or family members** who see the animation even though they do not own or operate the device.

The development organization initially declines to substantially alter the animation.

Possible fictional organizational rationales can include:

- the number of reports appears extremely small compared with the total user population;
- engineers cannot consistently reproduce the events;
- causal attribution is uncertain;
- reports are treated as anecdotal;
- the visual effect complies with an internal design specification;
- the animation is considered part of the intended visual identity;
- developers assume existing accessibility controls are sufficient;
- removing it would require engineering or design changes;
- no statistically obvious population-wide safety signal has yet appeared.

Treat these explanations as part of the simulation and **not as statements about Google's or Android developers' actual conduct.**

PART VII — EXPOSURE MODEL

Model risk as a chain rather than assuming every unlock carries equal seizure risk.

Consider stages such as:

Android population→ people exposed to the animation→ susceptible individuals→ visually/photosensitively susceptible individuals→ sufficient visual exposure→ triggering conditions present→ neurological response→ seizure event→ injury or other secondary consequence.

Account for repeated exposure.

For example, a hypothetical susceptible person seeing an animation 100 times per day experiences approximately:

100 exposures/day× 365 days= 36,500 exposures/year.

Over five years, absent behavioral changes, this becomes approximately:

182,500 exposures.

Explore whether repeated exposures change the probability model.

Explicitly distinguish between:

- probability per exposure,
- probability per person per year,
- probability of at least one event over repeated exposures.

Avoid simply multiplying probabilities where independence assumptions would be inappropriate.

PART VIII — HYPOTHETICAL ADVERSE-EVENT RATES

Because no causal event rate should be invented and presented as established fact, construct multiple fictional rates purely for simulation.

Include extremely rare through deliberately high stress-test values, for example:

- 1 adverse event per 10 billion exposures,
- 1 per billion,
- 1 per 100 million,
- 1 per 10 million,
- 1 per million,

along with alternative models in which risk is concentrated almost entirely among susceptible individuals rather than uniformly distributed across the population.

Clearly mark all such values as **experimental assumptions**.

For each scenario calculate:

- expected number of events,
- expected number of affected individuals,
- uncertainty,
- expected events per year,
- cumulative five-year events.

Where appropriate, use Poisson or binomial modeling rather than treating expected values as guaranteed event counts.

PART IX — REPRESENTATIVE-SAMPLE QUESTION

Determine how likely a randomly selected person would be to personally encounter someone who experiences such an event.

Compare several environments.

Scenario A — General population

A person whose social and occupational contacts roughly resemble the general population.

Scenario B — Low-contact individual

Someone who interacts with relatively few different people.

Scenario C — High-contact occupation

Someone regularly interacting with hundreds or thousands of people.

Scenario D — Disability and support-service environment

A worker whose daily contacts are **not a representative sample of the general population** and who works with individuals with physical disabilities, developmental disabilities, chronic medical conditions, neurological conditions, or other health needs.

Do not assume that every person with a disability has elevated seizure risk.

Instead, model different hypothetical prevalence levels to determine how **population composition changes encounter probability**.

For example, compare environments with epilepsy prevalence equal to:

- general-population baseline,

- 2× baseline,
- 3× baseline,
- 5× baseline,

only as sensitivity analyses unless empirical evidence supports a particular value.
Calculate the probability that a worker interacting with N individuals encounters:

- at least one person with epilepsy,
- at least one person with photosensitive epilepsy,
- at least one susceptible person,
- at least one adverse event over a specified observation period.

PART X — FAMILY AND BYSTANDER EXPOSURE

The device owner should not be treated as the only exposed person.
Simulate situations involving:

- spouses,
- children,
- parents,
- roommates,
- coworkers,
- clients,
- caregivers,
- passengers,
- people sitting nearby.

Estimate how secondary exposure changes the total number of visual exposures.
Model cases where the susceptible person is **not the Android user**.

PART XI — OUTCOME SEVERITY

Not every event should be assigned the same consequence.
Create an outcome tree including hypothetical categories such as:

1. No adverse response.
2. Momentary visual discomfort.
3. Headache or migraine.

4. Dizziness/disorientation.
5. Aura or neurological symptoms without seizure.
6. Seizure without secondary injury.
7. Seizure requiring medical evaluation.
8. Seizure producing secondary injury from falling or collision.
9. Emergency-department treatment.
10. Hospitalization.
11. Rare severe secondary outcome.

Do not artificially assign percentages to medical outcomes without either evidence or clearly labeling them as sensitivity assumptions.

PART XII — DETECTION PROBLEM

Examine an important systems issue:

Could a real adverse effect exist yet remain difficult to detect?

Model underreporting.

For example, if only:

- 100%,
- 50%,
- 25%,
- 10%,
- 5%,
- 1%,

of actual events are reported to the platform developer, calculate how the apparent incident rate changes.

Include additional losses caused by:

- users not connecting the event to the animation,
- users never filing reports,
- reports being categorized incorrectly,
- medical events not being communicated to the developer,
- family members not knowing what caused the exposure,
- duplicate reports,
- false attribution.

Distinguish **signal detection** from actual causal proof.

PART XIII — ENGINEERING RESPONSE SIMULATION

Simulate several developer policies.

Policy 1 — No change

Animation remains mandatory.

Policy 2 — Accessibility opt-out

Users can completely disable the effect.

Policy 3 — Reduced-motion integration

The animation automatically changes when "Reduce Motion" or a comparable accessibility feature is enabled.

Policy 4 — Safer redesign

The animation remains but eliminates problematic luminance/flash characteristics.

Policy 5 — Automatic risk mitigation

The system defaults to the safer animation and lets users voluntarily enable the more elaborate visual effect.

Calculate how each policy changes cumulative exposure and modeled events.

This section should emphasize the difference between:

- prevalence of harm,
- severity of harm,
- avoidability of harm,
- engineering cost,
- accessibility principles.

A very rare effect may still deserve mitigation when the exposure is unnecessary and technically avoidable.

PART XIV — MONTE CARLO SIMULATION

Build a reproducible simulation.

Use a synthetic population containing variables such as:

- age,

- Android use,
- daily unlock frequency,
- seizure disorder status,
- photosensitivity status,
- exposure to other people's devices,
- susceptibility state,
- individual triggering threshold,
- animation exposure count,
- adverse response,
- seizure occurrence,
- outcome severity,
- report/no-report status.

Run sufficient iterations to estimate distributions rather than only single expected values.
For each scenario report:

- mean outcome,
- median,
- 5th–95th percentile,
- probability of zero events,
- probability of ≥ 1 event,
- probability of ≥ 10 events where relevant,
- population incidence.

Run sensitivity analysis across uncertain parameters.

PART XV — SAMPLE-SIZE ANALYSIS

Answer:

If the animation really produced a very rare adverse event, how large would a study have to be before researchers would reasonably expect to observe one?

Calculate the approximate sample or exposure count required to have:

- 50% probability,
- 90% probability,
- 95% probability,
- 99% probability

of observing at least one event under several hypothetical true event rates.

This should demonstrate why rare adverse events may not appear during small usability studies.

PART XVI — PERSONAL-NETWORK VERSUS POPULATION RISK

Explicitly explain the distinction between:

Population probability

The probability of an event across billions of people and billions/trillions of exposures.
and

Personal observational probability

The probability that one particular person will ever directly witness such an event.

Demonstrate mathematically why:

"I have never seen this happen"

does not necessarily imply:

"This does not happen."

Also demonstrate the converse:

Observing one temporally associated event does not establish that the animation
caused it.

Include base-rate reasoning and causal-inference limitations.

PART XVII — FIVE-YEAR PLATFORM SIMULATION

Run the fictional Android system for five years.

For every simulated year report:

- Android user population,
- total unlocks,
- total animation exposures,
- susceptible population,
- predicted adverse responses,
- predicted seizures under each assumed risk model,
- reports reaching the developer,
- serious outcomes,
- cumulative outcomes.

Compare:

1. mandatory animation for five years;

2. removal after Year 1;
 3. accessibility opt-out after Year 1;
 4. safer redesign after Year 1;
 5. immediate safer-by-default design.
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PART XVIII — ADVERSARIAL REVIEW

Challenge the simulation itself.

Identify assumptions capable of exaggerating the risk, including:

- treating phone pickups as unlocks,
- overestimating photosensitivity,
- assuming every flash-like stimulus is capable of triggering seizures,
- assuming independence among repeated exposures,
- assuming correlation proves causation.

Then identify assumptions capable of **underestimating** risk, including:

- ignoring bystanders,
- ignoring underreporting,
- ignoring unusually high-frequency users,
- excluding vulnerable subpopulations,
- assuming accessibility settings eliminate every visual effect,
- considering only medically confirmed seizures.

Produce both a strongest skeptical interpretation and a strongest precautionary interpretation.
Do not force either position to win.

FINAL QUESTIONS

Conclude by answering:

1. Approximately how many people worldwide currently live with epilepsy?
2. How has that estimate changed over the preceding five years?
3. What could it plausibly look like five years from now?
4. What proportion may be vulnerable to visually induced seizures?
5. How frequently does a typical smartphone user unlock or activate their device?
6. How different is exposure for light versus extremely heavy users?
7. At Android scale, how many animation exposures could occur each year?
8. How rare could an adverse event be and still produce observable cases at global scale?

9. How likely would a conventional usability study be to detect an extremely rare event?
 10. How likely is one ordinary person to personally encounter such an event?
 11. How does that probability change for someone working with a nonrepresentative population containing higher concentrations of medical or neurological conditions?
 12. Could family members or bystanders materially increase exposure?
 13. How strongly would underreporting obscure a real signal?
 14. How much risk reduction would an accessibility opt-out create?
 15. Would removing an unnecessary repeated exposure substantially change modeled outcomes even if the underlying event probability were extremely small?
 16. What evidence would actually be necessary before concluding that an unlock animation caused seizures?
 17. What evidence would be necessary before reasonably ruling the hypothesis out?
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OUTPUT REQUIREMENTS

Provide:

- a research summary;
- source-supported prevalence data;
- a five-year historical table;
- a five-year projection;
- Android population assumptions;
- phone-unlock frequency ranges;
- exposure calculations;
- representative versus nonrepresentative population comparisons;
- probability calculations;
- Monte Carlo simulation;
- sensitivity analysis;
- developer-response scenarios;
- five-year cumulative results;
- adverse-outcome model;
- limitations;
- adversarial review;
- conclusions.

Every number must be classified as one of:

Observed / Published Estimate / Derived Calculation / Projection / Hypothetical Assumption / Simulation Result.

Do not allow simulated numbers to silently become factual medical statistics.

The analysis should be capable of discovering either:

- no meaningful population-level hazard,

- a hazard too small to distinguish from background events,
- a small but detectable susceptible-subpopulation effect,
- or a sufficiently strong signal that engineering mitigation would materially reduce risk.

The simulation must therefore be designed to **test the proposition rather than prove it**.

One aspect I particularly want to preserve from your original question is the distinction between a **representative population and the population a particular person actually encounters**.

Someone could have relatively few daily contacts while simultaneously working in an environment whose health characteristics differ substantially from the general public. Those are two different statistical variables, and we can model both without assuming that disability itself implies seizure susceptibility.

The next stage can use current WHO/peer-reviewed epilepsy and photosensitivity estimates, current Android/user estimates, smartphone-interaction research, and then an actual Monte Carlo simulation. That would let us put real values around the particularly interesting question here: **how can an event be extraordinarily unlikely per individual exposure but become nontrivial after billions or trillions of repeated exposures?**

CLAUDE AI

Here's the detailed rewrite:

Objective Build a research-style study, partly real-data and partly fictional, on how a hypothetical non-optional flash-like Android unlock animation could affect people with seizure disorders. The study should also model how likely harm would be in a population like the one I work with, compared with a general sample.

Part 1: Baseline epidemiology (real data)

- Estimate how many people worldwide have a seizure disorder, as a global average and as a rate per 1,000 people. Distinguish epilepsy overall from photosensitive epilepsy, the subset triggered by flashing light.
- Compare the past five years (2021–2026) with a five-year projection (2026–2031), stating data sources, assumptions, and uncertainty ranges.

Part 2: Exposure estimate (real data plus assumptions)

- Estimate how often people unlock their phones per day, from light to average to heavy users, and convert that to unlocks per year.
- Estimate how many unlocks per year would involve the flash effect, for each user tier.

Part 3: Fictional scenario for study

- Premise: Android ships a flash-like unlock animation with no opt-out.
- Define the animation's characteristics: flash frequency, brightness, duration, color contrast.
- Model what share of users, and of family members who share devices or are present, might have a seizure because of it.
- Describe outcomes: injuries, emergency visits, long-term effects, and effects on trust.

- Include a fictional developer response: unwillingness to acknowledge or address the risk for a vulnerable subset. Show how that choice affects outcomes over time.

Part 4: Simulation

- Design a simulation lab that tests this scenario, with explicit rules, variables, and starting conditions.
- Run it across many trials and report the probability of at least one seizure event in a representative sample, with confidence ranges.

Part 5: My population versus a representative sample

- Re-run the model for a group like the people I serve: individuals with mental and physical disabilities and chronic health conditions, where seizure disorders and photosensitivity are likely more common.
- Compare the two results and explain why my daily experience of the risk differs from the general population's.

Output preferences

- Single-column, mobile-readable format with no tables.
- State assumptions and mark which numbers are real and which are fictional or modeled.

If this captures what you want, I'll start with Part 1 and work through in order. Anything to add or change first?

PERPLEXITY AI

Rewritten prompt

Create a clearly labeled, fictional public-health and accessibility risk simulation involving a hypothetical Android phone unlock animation that includes a mandatory, bright flashing visual effect. The effect cannot be disabled, reduced, or replaced through accessibility settings.

The analysis should:

1. Estimate how many people worldwide live with epilepsy or another seizure disorder, using the best available global prevalence estimates and clearly distinguishing:
 - People living with epilepsy or active seizure disorders.
 - People newly diagnosed each year.
 - The smaller subgroup with photosensitive epilepsy or other conditions for whom flashing or flickering visual stimuli may be a particular trigger.
 - People without epilepsy who could still experience other adverse effects, such as migraine, dizziness, sensory overload, panic, or sleep disruption.

2. Compare estimated worldwide epilepsy prevalence across the previous five years, using actual historical estimates where available. If annual global figures are not available, explain the limitation and use a transparent, evidence-informed interpolation or scenario range rather than presenting invented precision.
3. Project a five-year future scenario using multiple assumptions:
 - A low-growth scenario.
 - A population-growth-only scenario.
 - A higher-burden scenario that considers aging, improved diagnosis, unequal access to care, and population growth.
 - Clearly label every future number as a scenario estimate, not a prediction.
4. Estimate smartphone unlock exposure under several behavioral categories:
 - Light user.
 - Typical user.
 - Frequent/heavy user.
 - Very heavy or compulsive-use user.
 - Include calculations for unlocks per day, month, and year.
 - Explain that an unlock count is not necessarily the same as the number of flash exposures, since an animation might not play in every unlocking pathway.
5. Build a hypothetical Android-user exposure model in which the unlock animation has a flash or strobe-like effect and users cannot opt out. Include assumptions about:
 - Android user population.
 - Device adoption and replacement rate.
 - Share of users exposed to the animation.
 - Daily unlock frequency.
 - Probability that the animation produces a visually hazardous flash pattern.
 - Probability that a susceptible person experiences a seizure after an exposure or series of exposures.
 - Probability of secondary harms, such as falls, injuries, anxiety, avoidance of phone use, caregiver burden, emergency medical visits, or loss of independence.
6. Construct several fictional study scenarios, including:
 - A baseline scenario with no elevated risk.
 - A low-risk scenario.
 - A moderate-risk scenario.
 - A high-risk or design-failure scenario.
 - A scenario in which the animation is more likely to affect users with photosensitive epilepsy, migraine sensitivity, traumatic brain injury, autism-related sensory sensitivity, intellectual/developmental disabilities, visual processing differences, medication-related vulnerabilities, or other chronic health conditions.
 - A household or family-member exposure scenario, recognizing that a phone screen can affect someone nearby even if they are not the device owner.
7. Estimate expected outcomes in a representative sample, such as 10,000, 100,000, and 1,000,000 Android users. For each scenario, calculate:
 - Number of users exposed.
 - Number of users in a potentially susceptible subgroup.

- Number of potential triggered seizures under stated assumptions.
 - Number of potential family-member or bystander events.
 - Expected healthcare, injury, quality-of-life, accessibility, and trust impacts.
 - Uncertainty ranges and the assumptions driving them.
8. Include a separate “non-representative setting” analysis for disability-support, day-program, residential-care, and community-service environments. Address why a group composed of people with intellectual/developmental disabilities, physical disabilities, chronic medical conditions, neurological disorders, medication use, communication barriers, and sensory vulnerabilities may have a substantially different risk profile than the general public. Do not assume that the general-population average applies to that setting.
 9. Analyze the ethical, accessibility, product-design, and organizational implications if Android developers were aware of credible risk signals but declined to offer an opt-out, reduced-motion mode, flash-safety setting, or other mitigation. Discuss:
 - Foreseeability of harm.
 - Accessibility-by-design.
 - Informed choice and user autonomy.
 - The consequences of forcing exposure through a basic device function such as unlocking.
 - The need for incident reporting, testing, independent review, and rapid rollback mechanisms.
 - The difference between a fictional policy-analysis scenario and a factual allegation about any real company or product.
 10. Propose a safe simulation methodology rather than testing flashing effects on people. Use computer modeling, literature-based parameters, anonymized incident reports, accessibility review, device telemetry with consent, synthetic populations, and sensitivity analyses. Do not recommend deliberately exposing seizure-prone participants to potentially unsafe flashing stimuli.
 11. Present the result as a structured report with:
 - Definitions and scope.
 - Evidence base and limitations.
 - Assumptions table.
 - Five-year retrospective prevalence comparison.
 - Five-year scenario projection.
 - Unlock-frequency exposure calculations.
 - Fictional risk scenarios.
 - Representative-sample simulations.
 - Disability-support-setting simulation.
 - Ethical and developer-responsibility analysis.
 - Risk mitigations and accessibility recommendations.
 - A clear statement that the scenario is fictional and not evidence that a real Android unlock animation causes seizures.

Important framing correction

“Seizure disorder” can mean several things, so the most defensible global proxy is **epilepsy**, especially “active epilepsy” or epilepsy requiring ongoing treatment. The World Health Organization estimates that around **50 million** people worldwide have epilepsy, with active epilepsy estimated at about **4–10 per 1,000 people**; it also estimates about **5 million new diagnoses annually**. A Global Burden of Disease analysis estimated **51.7 million people** living with combined idiopathic and secondary epilepsy in 2021, or about **0.7% of the global population**.[\[who\]](#)[\[thelancet\]](#)

That number should **not** be treated as the number of people who are vulnerable to flash-triggered seizures. Photosensitive epilepsy is a narrower subgroup, and the appropriate rate varies by age, epilepsy type, diagnostic practice, and country. A serious analysis should model it as a range, not assume that every person with epilepsy has the same sensitivity.

Five-year prevalence comparison

There is no single, clean, globally standardized annual count for every year that can be treated as a census. The strongest global benchmark returned by the search is the GBD 2021 estimate of 51.7 million people living with epilepsy, with a 95% uncertainty interval of 44.9–58.9 million. WHO’s rounded public-health estimate is approximately 50 million.[\[who\]](#)[\[thelancet\]](#)

For a planning model, use the following **illustrative reconstruction**, not as a claim of exact annual surveillance:

Year	Illustrative global epilepsy prevalence	Interpretation
2021	51.7 million	GBD benchmark estimate
2022	52.1 million	Scenario interpolation based on population growth and stable prevalence
2023	52.5 million	Scenario interpolation, not direct measurement
2024	52.9 million	Scenario interpolation, not direct measurement
2025	53.3 million	Scenario interpolation, not direct measurement

2026	53.7 million	Current-year scenario estimate, not a confirmed global count
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This illustrative series assumes that the absolute number rises about 0.7% annually, broadly consistent with population growth and aging while not asserting that the underlying rate itself necessarily rises. The 2021 GBD analysis found that the absolute burden had risen over the long term, while age-standardized patterns differ by epilepsy category and location. [\[thelancet\]](#)[\[healthdata\]](#)

Five-year forward scenarios

The table below treats 2026 as an illustrative baseline of 53.7 million people living with epilepsy. It is a planning exercise, not a clinical forecast.

Scenario	Annual change assumption	2031 illustrative total	What it represents
Lower-growth	0.3% per year	54.5 million	Better prevention and treatment access offset much of population growth
Population-growth-only	0.7% per year	55.6 million	Prevalence rate stays broadly stable while global population grows
Higher-burden	1.2% per year	57.0 million	Population aging, improved recognition, survival, and unequal care access raise absolute burden

These scenarios are deliberately conservative in how they are described. They do **not** establish that epilepsy will increase by any particular amount. They illustrate why a product feature deployed at operating-system scale can reach a large number of people even when the individual risk is uncommon.

Unlock exposure model

Published estimates of smartphone checking and unlocking vary widely because researchers measure different things: pickups, checks, screen activations, and unlocks. One frequently cited

consumer-data range puts unlocking at roughly **96–150 times per day**, while a recent user-reported tracking example found 44 daily unlocks. Those figures are useful as behavioral reference points but are not a clinical or globally representative standard.[\[slicktxt\]](#)[\[reddit\]](#) For a transparent simulation, use tiers rather than pretending there is one universal average:

User category	Assumed unlocks/day	Unlocks/month	Unlocks/year
Light user	25	750	9,125
Typical user	60	1,800	21,900
Frequent user	120	3,600	43,800
Heavy user	200	6,000	73,000
Very heavy user	300	9,000	109,500

If every unlock caused one flash exposure, a person unlocking 120 times daily would encounter approximately **43,800 flash exposures annually**. If the effect appeared only after a certain kind of unlock, say 50% of unlocks, the exposure estimate becomes approximately **21,900 yearly exposures**.

This distinction matters: rare per-exposure risks can accumulate across thousands of exposures. But the risk cannot be estimated responsibly without knowing the animation’s brightness, flash frequency, contrast, duration, screen size, viewing distance, ambient light, and whether it meets recognized flash-safety thresholds.

Fictional scenario

This is a **fictional model**, not an allegation that a real Android animation has caused seizures. Assume an Android platform update introduces a full-screen, high-contrast unlock animation with rapid luminance changes. The animation is mandatory: it cannot be disabled through accessibility settings, reduced-motion settings, developer options, or parental controls. Assume a synthetic population of 1,000,000 Android users:

Model input	Illustrative assumption
Users receiving the feature	90% or 900,000 users
Average unlocks among exposed users	100 daily

Animation shown on eligible unlocks	80%
Annual exposures per affected user	29,200
People with epilepsy	0.7%, or 7,000 users
Narrow flash-sensitive subgroup	0.05%–0.3%, or 500–3,000 users
Nearby family/bystanders regularly exposed	20% of users, or 200,000 people
Design control	No opt-out, no reduced-motion mode, no warning, no rapid rollback

At 100 daily unlocks with the animation appearing on 80% of them, the 900,000 affected users experience:

$$900,000 \times 100 \times 0.80 = 72,000,000$$

potential animation exposures per day, or approximately:

$$72,000,000 \times 365 = 26.28 \text{ billion}$$

potential exposures per year.

The presence of billions of exposures does **not** mean billions of harmful events. It means that even a very low individual-event probability could become material at population scale—particularly if the design is unsafe for a known susceptible subgroup.

Illustrative outcome simulation

Below, “trigger probability” means the fictional probability that a vulnerable person experiences a seizure associated with an exposure period. It is **not** a real-world estimate and should never be used as a medical risk figure.

Scenario	Flash-sensitive users in 1M-user sample	Assumed annual event probability per susceptible user	Expected directly affected users/year	Possible bystander/family events

Baseline / safe design	500–3,000	Near zero because the animation meets safety standards or is disabled	0–1	0
Low-risk design failure	500–3,000	0.05%	0–2	0–1
Moderate-risk design failure	500–3,000	0.5%	3–15	1–5
Severe design failure	500–3,000	2%	10–60	3–20

The expected-event calculation is:

$\text{Expected events} =$

$\text{Susceptible users} \times$

$\text{annual event probability}$

For example, if 3,000 users are in the susceptible subgroup and the hypothetical annual event probability is 0.5%:

$3,000 \times 0.005 = 15$

expected directly affected users in the year.

A family-member or bystander event would typically be modeled separately, with a much lower exposure rate. Yet it should not be ignored: a phone may be used in a car, bedroom, shared living room, day program, group home, or crowded community setting. A flashing effect can affect a nearby person even when they never chose the phone, update, or animation.

Why representative samples may mislead

A general-population sample is useful for estimating broad deployment impact, but it can be misleading in the setting you described. In a support program, day program, group setting, residential service, or disability-focused community environment, the population may differ from the general public in clinically relevant ways.

That does **not** mean every disabled person has elevated seizure risk, and it would be inappropriate to make that assumption. But the setting may have a higher concentration of people who have epilepsy, neurological conditions, traumatic brain injury, medication effects, migraines, sensory-processing differences, communication barriers, mobility limitations, fall risk, or difficulty independently changing device settings.

In such a setting, a harmful unlock animation could produce more than a direct seizure count:

- A seizure or near-seizure event could lead to falls, aspiration risk, injury, emergency evaluation, missed programming, and temporary loss of independence.

- A non-seizure sensory reaction could cause distress, dysregulation, panic, behavioral escalation, migraine, nausea, or avoidance of an essential communication device.
- Staff could face a more complex safety burden: monitoring visual triggers, changing workflows, documenting incidents, reassuring family, and finding workarounds for a feature users cannot disable.
- A person who relies on an Android phone for communication, transportation, health reminders, scheduling, or assistive technology may be harmed if they avoid unlocking it because they fear the animation.

A more appropriate simulation for a disability-support program would use local, anonymized aggregate characteristics—such as the number of people with documented seizure plans, known light sensitivity, migraines, relevant medication profiles, visual sensitivities, and fall-risk precautions—rather than importing a 0.7% global average.

Fictional developer-response scenario

In the fictional case, Android developers receive reports that the animation may involve rapid flashes, bright transitions, and high contrast. Some reports describe seizures; others report migraine, disorientation, sensory distress, or avoidance. The developers conclude that the effect is visually distinctive and decide not to remove it, make it optional, or connect it to reduced-motion controls.

The core ethical problem is not merely whether every report can be proven causal. It is that the feature is:

- Repeatedly encountered through a basic device function.
- Difficult or impossible to avoid without abandoning normal phone use.
- Potentially foreseeable as a problem for a known medically vulnerable subgroup.
- Easy to mitigate through design changes, such as no-flash transitions, reduced motion, user-controlled themes, brightness limits, and a complete opt-out.
- Being imposed on people who may depend on the phone as an accessibility or communication tool.

In a real organization, the responsible response would be to investigate promptly, provide a server-side or app update rollback when possible, add an immediate opt-out, assess the animation against visual-flash safety guidance, conduct independent accessibility review, and transparently communicate the issue. “No confirmed causal proof yet” is not a sufficient reason to force ongoing exposure when the potential harm is severe and the mitigation is low cost.

Safe simulation design

Do not test a potentially hazardous flashing animation by exposing people with known or suspected photosensitivity. A safer study can be designed around computer modeling and observational data.

1. **Define the visual stimulus.** Measure luminance change, contrast, flash frequency, duration, full-screen coverage, color transitions, and whether the display behavior changes by device model or refresh rate.
2. **Build a synthetic population.** Create simulated users by age range, device use tier, known epilepsy prevalence, a narrower photosensitivity range, migraine/sensory-sensitivity ranges, and household exposure patterns.
3. **Model exposure.** Assign each user a daily unlock frequency and percentage of unlocks where the animation appears. Run the simulation across a year to estimate cumulative exposures.
4. **Use sensitivity analysis.** Vary each uncertain parameter across low, medium, and high values. This reveals whether conclusions depend on one unverified assumption.
5. **Use passive safety evidence.** Analyze anonymized support tickets, voluntary incident reports, accessibility feedback, clinical reports where ethically and legally appropriate, and changes in migraine/seizure-related complaints following feature rollout.
6. **Compare against a control period.** If a feature is released, compare incident reporting before and after rollout, while accounting for device adoption, media attention, reporting bias, and unrelated seizure variation.
7. **Create a stop rule.** Predefine thresholds for pausing rollout, disabling the feature, or adding an opt-out if credible safety signals emerge.
8. **Include disability-service settings separately.** Model a general public sample and a high-support-needs setting independently. Do not dilute the latter's safety signal by combining it into a much larger low-risk population.

Bottom line

Around **50–52 million people globally** were estimated to be living with epilepsy in recent global assessments, but only a subset would be expected to have flash-sensitive seizure vulnerability. A forced flash-based Android unlock animation is therefore best analyzed as an accessibility and risk-management problem: each user may see it tens of thousands of times annually, so a low-probability harm can become meaningful when multiplied across a very large user base. [\[who\]](#) [\[thelancet\]](#)

For a general population, the absolute risk may appear small under many plausible assumptions. For a disability-support environment with a higher-than-average concentration of people with neurological, medical, sensory, and mobility vulnerabilities, the average global risk can be the wrong comparison entirely. The appropriate design principle is simple: a repetitive visual effect on an essential device function should be optional, compatible with reduced-motion and accessibility settings, and rapidly reversible when credible safety concerns arise.

GROK AI

Rewritten prompt, then the numbers, then a fictional study you can actually rerun. This is not a finding that any current Android unlock animation causes seizures.

Rewritten prompt

Build a bounded research brief and a testable simulation on three linked questions.

First, estimate how many people worldwide have a seizure disorder, meaning active epilepsy under the International League Against Epilepsy sense of an enduring predisposition to unprovoked seizures. Compare the last five years with a five-year forward band. Separate idiopathic from secondary epilepsy. State uncertainty. Do not treat a single prevalence figure as a headcount of people who will seize if they see a light.

Second, estimate daily phone unlocks for light, standard, and heavy users, and say which figures are logged and which are self-report.

Third, write a fictional product scenario, clearly labeled fiction, in which an Android unlock animation is a flash or wave of light, ships as a non-optional system transition, and the platform owner declines to treat it as a photosensitivity question. Specify the stimulus in physical terms: number of luminance reversals, duration, contrast, and whether it meets or misses the WCAG 2.3.1 three-flashes threshold. Then run a study design in which a parameterized share of users, and a smaller share of nearby household members, have a seizure attributed to that animation. Include outcomes, reporting failure, and the developer's refusal to add an opt-out or to measure the animation against photic-stimulation criteria.

The simulation must not assume the animation is a 3–30 Hz strobe. It must sweep hypothetical per-unlock hazard rates, including zero. It must compare a representative Android sample with a non-representative direct-support caseload of people with intellectual, developmental, physical, and chronic health disabilities, where epilepsy is far more common but photosensitive epilepsy is still a small slice of that. Report how often such a caseload would be expected to observe the outcome in a year. Draft only. Not medical advice. Not a claim about any shipping phone.

How many people have a seizure disorder

The best current headcount is the Global Burden of Disease 2021 analysis, published in 2025: **51.7 million people with active epilepsy** (95% uncertainty interval 44.9–58.9 million). That is about **0.7% of the 2021 population**. Idiopathic epilepsy was 24.2 million. Secondary epilepsy was about 27.5 million. Age-standardized prevalence was 658 per 100,000.

A 2026 meta-analysis of 216 studies pooled prevalence at **7.57 per 1,000** (95% CI 6.78–8.45), which lands in the same band.

Last five years, without inventing a yearly table the GBD did not publish:

- The prior GBD round, data through 2016, counted **45.9 million**. The jump to 51.7 million by 2021 is population growth, more secondary epilepsy, and some method change, not a sudden epidemic.
- Age-standardized prevalence rose **10.8% from 1990 to 2021**, mostly from secondary epilepsy. From 2010 to 2021 the GBD estimated annual change was not significant.
- So 2021–2026 is best read as a roughly flat rate on a slowly growing, aging population. Holding the 2021 crude rate near 0.65% and applying population growth puts **2026 around 53–55 million**. That is an extrapolation, not a new GBD year.

Five years forward, same method: if the rate stays flat and world population is near 8.5–8.6 billion by 2031, the count is about **55–58 million**. If secondary epilepsy keeps rising with stroke, dementia, trauma, and infection, the count can sit higher while the idiopathic rate stays flat. Deaths and disability rates for idiopathic epilepsy have been falling even as prevalence counts rise.

Photosensitive epilepsy is the only slice relevant to a light trigger. About **3% of people with epilepsy**, with sources spanning **2–5%**. Epilepsy Action puts the population rate near **1 in 4,000**. Peak laboratory triggers are **15–25 flashes per second**, with a wider sensitive band of about **3–60 Hz**. Most onset is between ages 7 and 19.

On an 8.25 billion population that is roughly **1.6–2.6 million** people with photosensitive epilepsy worldwide. It is not 51.7 million.

Unlocks per day

Logged and self-report figures disagree, and they should not be averaged into one “true” number.

- 2018 U.S. survey by generation: Silent generation 18, boomers 30, Gen X 49, millennials 63, Gen Z 79.
- Logged college-phone study, 2017–2022: most students **50–100 unlocks** on an ordinary day, **100–300** on a high-use day, sessions mostly under 4 minutes.
- Logged U.S. Android teenagers, 2022: median **51 pickups**, range 2 to 498.
- Late-2025 consumer survey: average American **186 reaches** a day. That is self-report and a weaker figure.

Working bands for the simulation: light **30**, standard logged **60**, standard self-report **120**, heavy **200**, very heavy **300**. At 60 unlocks a day that is about **22,000 unlocks a year**. At 200 it is about **73,000**.

Android is the relevant installed base for this scenario: about **72% of mobile devices, roughly 3.9 billion active users**.

The stimulus caveat, before the fiction

WCAG 2.3.1 fails content that flashes **more than three times in any one second**, above a defined luminance threshold. A general flash is a pair of opposing luminance changes, not a single brightening.

A one-shot unlock wave is one transition, or at most one pair, spread over a few hundred milliseconds. That is **not** a 15–25 Hz strobe. Under published photic-stimulation criteria the seizure hazard of that stimulus is the zero arm of the model, unless measurement shows it actually reverses more than three times in a second or carries a high-contrast pattern.

Discomfort, startle, migraine, and motion sensitivity are separate problems and do not require a seizure disorder.

Fictional scenario

Label: fiction. Not a description of Pixel, Android 16, or any shipping build.

In 2027 a platform group ships “Lumen Wake,” a lock-to-unlock signature: a full-screen white wave, peak contrast, about 400 ms, no reduced-motion hook, no setting. Brand review calls an opt-out fragmentation. Accessibility review checks TalkBack and contrast, not flash count. Support macros file tickets as preference. Telemetry counts unlocks and does not count seizures, so “no safety signal in telemetry” is guaranteed by the instrument. After trade press asks, the public line is that a single transition is below the three-flash rule, which may be true, and that the team will not measure it against intermittent photic stimulation criteria or ship a plain cut.

Study arms, all fictional:

- Arm A: animation as shipped, no opt-out.
- Arm B: same animation, one-time off switch.
- Arm C: hard cut, no luminance wave.

Primary outcome is a clinical seizure within 30 seconds of an unlock, confirmed by witness or video, in someone with a prior photosensitive diagnosis or a positive photoparoxysmal EEG. Secondary outcomes are EEG change without a seizure, migraine, drop attack from startle, fall injury, and household exposure when the phone is unlocked in a shared room. Bystander dose is lower because the screen fills less of their visual field.

Pre-registered attribution rule: an event counts only if the animation measured above the flash threshold, or the person already had a positive photic EEG and the event was time-locked. Otherwise it is recorded as coincidence. The fictional developer rejects that rule and asks for proof that the animation, rather than sleep loss or missed medication, was the cause, while refusing the measurement that would settle it.

Simulation

Anchors used: 3.9 billion Android users, photosensitive rate 1 in 4,000, so about **975,000** Android users with photosensitive epilepsy. Hazard is a per-unlock probability applied only to that group. Zero is included on purpose.

At 60 unlocks a day, expected events in a year among those 975,000 people:

- $p = 0$: **0**
- $p = 1$ in 10 million unlocks: about **2,100** events, and about a **0.2%** chance a given photosensitive user has one in a year
- $p = 1$ in 1 million: about **21,000** events, about a **2%** individual annual chance
- $p = 1$ in 100,000: about **214,000** events, about a **20%** individual annual chance

Those higher arms would already be obvious in epilepsy clinics. They are useful as ceilings, not as realistic guesses for a single wave.

Your work setting is the other distribution. In general samples of people with intellectual disability, pooled epilepsy prevalence is about **22%**, and it rises with severity, versus about 1% in the general population. Photosensitive epilepsy is still only a few percent of that epilepsy.

On an illustrative caseload of 12:

- Expected people with epilepsy: about **2.7**
- Expected people with photosensitive epilepsy: about **0.08**

So the usual year contains no photosensitive person at all. A 20,000-trial Monte Carlo at 80 unlocks a day found a seizure in that caseload in about **0.2% of years** at $p = 1$ in 1 million, and about **2% of years** at $p = 1$ in 100,000. A representative sample of 1,000 Android adults is only modestly more informative at the low hazards, because 1,000 times 1/4,000 is still a quarter of a photosensitive person.

The last arm in the file treats every person with epilepsy as if they were photosensitive. That is the wrong model. It is there to show how fast the result inflates if the study confuses “has a seizure disorder” with “will seize from this light.”

Files:

In a representative sample, a true effect at the WCAG-consistent bound does not appear, because the bound is zero. A rare effect can still be real in the world and absent from a 12-person caseload for years. The caseload raises the chance of epilepsy, medication interactions, falls, and missed doses. It does not by itself turn a single unlock wave into a photic trigger. If someone in that setting is already photosensitive, the practical control is the one the fictional platform refuses: a plain transition, reduced motion that actually removes the wave, and a way to turn the signature off.

MICROSOFT COPILOT

I'll first turn your idea into a precise, research-ready prompt. I'll also separate observed evidence from hypothetical assumptions so the eventual simulation does not imply that Android currently has this feature or that developers have actually ignored such reports. This is a strong accessibility-risk research concept, but it needs clear boundaries between real epidemiological evidence, modeled assumptions, and fictional corporate behavior. The rewritten prompt below is designed to support a later statistical simulation without falsely suggesting that Android currently uses such an animation or that its developers have ignored a documented hazard.

Rewritten research prompt

Title

Modeling the Public-Health and Accessibility Risks of a Hypothetical, Non-Optional Flashing Smartphone Unlock Animation

Purpose

Develop an evidence-informed, fictional risk-assessment study examining the potential consequences of a hypothetical smartphone operating-system update that introduces a bright, flash-like animation whenever an Android device is unlocked.

In this fictional scenario:

- The animation is enabled by default.
- Users cannot disable, reduce, replace, or opt out of it.
- The animation may contain changes in brightness, contrast, color, or visual frequency that could affect photosensitive individuals.
- The animation appears during every successful device unlock.
- It may be visible not only to the device owner, but also to family members, coworkers, clients, caregivers, passengers, and nearby bystanders.
- The fictional development organization initially declines to modify the animation despite accessibility reports and warnings.
- This organizational response is a scenario variable and must not be presented as a claim about Google, Android developers, device manufacturers, or any real organization.

The study should estimate the probability, frequency, and severity of adverse outcomes while giving special consideration to environments containing higher-than-average concentrations of people with disabilities, neurological conditions, chronic illnesses, or other forms of medical vulnerability.

Part 1: Define the population at risk

Estimate the worldwide number and percentage of people living with:

1. Active epilepsy.
2. A diagnosed seizure disorder, where reliable data distinguish this from epilepsy.
3. Photosensitive epilepsy or visually provoked seizures.
4. Undiagnosed photosensitivity.
5. Non-seizure sensitivity to flashing or rapidly changing visual content, including headaches, nausea, dizziness, disorientation, migraine, or vestibular symptoms.

Do not treat the terms *seizure*, *seizure disorder*, *epilepsy*, and *photosensitive epilepsy* as interchangeable.

Use the best available international evidence and report:

- The source year.
- The clinical or epidemiological definition used.
- The central estimate.
- The uncertainty interval, if available.
- The percentage of the world population represented.
- Relevant differences by age, geographic region, income level, disability status, or access to diagnosis and treatment.

As an initial evidence anchor, a 2025 Global Burden of Disease analysis estimated 51.7 million people with epilepsy in 2021, with a 95% uncertainty interval of 44.9 to 58.9 million. It estimated that approximately 0.7% of the global population had active epilepsy. The World Health Organization separately describes epilepsy as affecting around 50 million people worldwide.

[\[thelancet.com\]](https://www.thelancet.com) [\[who.int\]](https://www.who.int)

A commonly cited estimate indicates that approximately 3% of people with epilepsy experience seizures triggered by particular flashing lights or visual patterns. The risk depends on properties such as flash frequency, brightness, contrast, viewing distance, wavelength, and whether the viewer's eyes are open or closed. This percentage should be used only as an evidence-informed input range, not as the probability that any single animation will produce a seizure. [\[epilepsy.com\]](http://epilepsy.com)

Part 2: Construct the historical comparison

Compare the global burden for the five most recent years for which defensible data or modeled estimates can be produced.

For each year, report:

- Estimated world population.
- Estimated number of people with active epilepsy.
- Estimated prevalence per 100,000 people.
- Estimated number of people with photosensitive epilepsy.
- Lower and upper uncertainty bounds.
- Whether the value is directly reported, interpolated, extrapolated, or simulated.

Do not describe interpolated annual values as observed measurements.

Where annual global epilepsy estimates are unavailable, create transparent modeled estimates using:

- Published prevalence rates.
- Population estimates.
- Reported epidemiological trends.
- Sensitivity analyses using stable, increasing, and decreasing prevalence assumptions.

Explain how changes in total case counts can result from population growth even when the prevalence rate remains stable.

Part 3: Produce a five-year projection

Project the same measures for the next five years.

Include at least three forecast cases:

1. **Stable-prevalence scenario:** The prevalence rate remains constant while population changes.
2. **Trend-continuation scenario:** Recent prevalence trends continue.
3. **Uncertainty scenario:** Rates vary within a defensible upper and lower range.

Clearly label every future number as a projection rather than a known outcome.

Where appropriate, compare the model with published forecasting research. Existing research has projected epilepsy trends beyond 2031, but predictions may apply only to particular epilepsy categories or age groups. For example, one study projects idiopathic epilepsy among people

aged 15 to 49 through 2035 and therefore should not be generalized to the entire world population without adjustment. [\[pmc.ncbi.nlm.nih.gov\]](https://pubmed.ncbi.nlm.nih.gov/)

Part 4: Define the hypothetical unlock animation

Create a fictional animation specification with adjustable exposure characteristics:

- Number of flashes per unlock.
- Flashes per second.
- Duration in milliseconds.
- Maximum and minimum luminance.
- Luminance contrast.
- Presence of saturated red.
- Percentage of the display occupied.
- Full-screen versus localized presentation.
- Viewing distance.
- Ambient-light conditions.
- Device-screen brightness.
- Whether the exposure is viewed with one or both eyes.
- Whether the animation repeats after a failed authentication attempt.
- Whether rapid consecutive unlock attempts restart the animation.
- Whether nearby people can see the display.

Create at least three exposure profiles:

- **Lower-hazard profile**
- **Borderline or ambiguous profile**
- **Higher-hazard profile**

Evaluate these profiles against recognized accessibility guidance. Flashing content faster than three times per second, combined with sufficient area and contrast, can cross important accessibility thresholds. Photosensitive responses are often associated with flashes in the approximate range of 5 to 30 flashes per second, although individual sensitivity varies.

[\[webaim.org\]](https://www.webaim.org/) [\[epilepsy.com\]](https://www.epilepsy.com/)

Do not actually reproduce or display a hazardous flashing animation as part of the study.

Part 5: Estimate phone-unlock exposure

Review empirical research measuring actual smartphone unlocks rather than relying exclusively on self-reported screen time.

Estimate daily unlock distributions for:

- Light users.
- Standard or moderate users.
- Frequent users.
- Heavy users.

- Very heavy or compulsive-use profiles.

For each group, estimate:

- Mean unlocks per day.
- Median unlocks per day.
- A plausible range.
- Unlocks per waking hour.
- Annual unlocks.
- Number of animation exposures if one animation occurs per unlock.
- Additional exposures caused by failed unlock attempts or repeated authentication.

Do not invent a universal “average unlock count” if the literature does not support one. Smartphone-unlock behavior can vary substantially within the same individual over time, and research populations such as university students may not represent the general public. One continuous-sensing study used 47 undergraduates across 3,290 person-days and reported substantial within-person variability. [\[researchsquare.com\]](https://www.researchsquare.com)

Part 6: Model primary and secondary exposure

Distinguish between:

- **Primary exposure:** The device owner sees the animation while unlocking the phone.
- **Secondary exposure:** A family member, coworker, client, caregiver, passenger, or bystander sees the animation.
- **Repeated environmental exposure:** Several people unlock devices in a shared setting.
- **Concentrated-setting exposure:** The user works in a disability-service, healthcare, residential-care, rehabilitation, special-education, or community-support environment.

Model the number of people potentially exposed per unlock using several assumptions, such as:

- Owner only.
- Owner plus an average fraction of one bystander.
- Household exposure.
- Workplace exposure.
- High-contact care or support environment.

Avoid double-counting the same bystander repeatedly when calculating the number of unique individuals exposed.

Part 7: Simulate adverse outcomes

Build a probabilistic simulation, preferably a Monte Carlo model, with user-adjustable inputs. The simulation should distinguish among:

1. Exposure to the animation.
2. Exposure by a susceptible person.

3. Exposure to a stimulus that crosses a hazardous visual threshold.
4. A physiological response.
5. A clinically recognizable seizure.
6. A seizure causing secondary injury.
7. A non-seizure adverse effect.
8. No observable adverse outcome.

Potential modeled outcomes may include:

- No reaction.
- Visual discomfort.
- Headache or migraine.
- Nausea or dizziness.
- Disorientation.
- Brief absence or myoclonic event.
- Focal or generalized seizure.
- Fall or collision.
- Loss of device control.
- Emergency evaluation.
- Hospitalization.
- Temporary inability to work or participate in activities.
- Psychological distress or avoidance of the device.
- Harm to a nearby family member or client.
- Accessibility complaint.
- Product withdrawal, redesign, warning, or litigation within the fictional scenario.

Do not assign a seizure probability merely by multiplying epilepsy prevalence by the number of unlocks. The model must account for:

- The proportion of people with photosensitivity.
- Whether the stimulus is actually capable of provoking a response.
- Individual seizure thresholds.
- Repeated exposure without an event.
- Medication and seizure control.
- Fatigue, stress, sleep deprivation, illness, and other susceptibility modifiers.
- Dependence between repeated exposures involving the same person.
- Differences between risk per exposure and cumulative risk across time.

Part 8: Create fictional risk scenarios

Model at least four fictional scenarios:

Scenario A: General-population user

A user has ordinary daily contact with a relatively small number of people, broadly resembling general-population prevalence.

Scenario B: Heavy phone user

A user unlocks their device very frequently, increasing cumulative exposure opportunities.

Scenario C: Household with a susceptible family member

The owner does not have photosensitive epilepsy, but a household member may be exposed repeatedly.

Scenario D: Disability- and chronic-health-service environment

A user works regularly with people who have mental disabilities, physical disabilities, neurological conditions, chronic illnesses, or multiple health vulnerabilities.

For Scenario D, do not assume that disability automatically means photosensitivity or epilepsy. Instead, allow the simulated prevalence to differ from the general population only when supported by evidence or explicitly identified as a hypothetical enrichment factor.

Use several enrichment levels, such as:

- General-population prevalence.
- Two times general-population prevalence.
- Five times general-population prevalence.
- A user-defined prevalence based on the actual population being studied.

The purpose is to demonstrate how a person's personal network may differ from a statistically representative sample.

Part 9: Compare representative and non-representative personal networks

Estimate how likely it is for at least one susceptible person to appear in a network of:

- 10 people.
- 25 people.
- 50 people.
- 100 people.
- 250 people.
- 1,000 people.

Begin with the binomial approximation:

$P(\text{at least one susceptible person}) = 1 - (1 - p)^n$
where:

- p is the assumed prevalence of susceptibility.
- n is the number of unique people in the network.

Then explain why this approximation may be inappropriate when:

- People in the network are not independent.
- The population is enriched for certain health conditions.
- One person is encountered repeatedly.

- Household or workplace characteristics cluster risk.
- Diagnosis is underreported.
- The population differs systematically from the general public.

Compare both representative and enriched-network scenarios.

Part 10: Model the fictional organizational response

Create a fictional timeline in which users, caregivers, clinicians, accessibility specialists, and advocacy organizations report possible adverse effects.

The fictional developer organization initially responds by:

- Treating reports as isolated anecdotes.
- Requesting more evidence before changing the design.
- Arguing that the animation passes internal testing.
- Declining to include an opt-out setting.
- Prioritizing visual branding or design consistency.
- Delaying investigation because the affected subgroup appears statistically small.
- Failing to test adequately with disabled and medically vulnerable participants.

This conduct must be described as a hypothetical study condition, not as a factual characterization of Android, Google, or real Android developers.

Test alternative organizational policies:

1. No action.
2. Warning only.
3. Reduced intensity.
4. Reduced flash frequency.
5. “Reduce motion” compatibility.
6. User opt-out.
7. Safe default with optional enhancement.
8. Immediate remote disablement.
9. Pre-release accessibility testing.
10. Independent medical and accessibility review.

Compare the projected number of exposures and adverse events under each policy.

Part 11: Quantify uncertainty and rarity

For every risk estimate, report:

- Expected value.
- Median outcome.
- 95% simulation interval.
- Probability of zero events.
- Probability of one or more events.

- Probability of a severe event.
- Number needed to expose for one modeled event, if meaningful.
- Sensitivity to the assumed photosensitivity prevalence.
- Sensitivity to unlock frequency.
- Sensitivity to animation properties.
- Sensitivity to secondary exposure.
- Sensitivity to enriched personal networks.

Include rare-event methods when the expected event probability is very low. Run enough simulation iterations to stabilize tail estimates, and report when an event remains too rare to estimate reliably.

Part 12: Required study outputs

Produce:

1. An executive summary.
2. A five-year retrospective table.
3. A five-year prospective table.
4. A table of smartphone-use categories and unlock frequencies.
5. A table separating empirical inputs from fictional assumptions.
6. A causal pathway diagram.
7. A representative-network probability analysis.
8. An enriched-network probability analysis.
9. Monte Carlo simulation results.
10. Best-case, central, and worst-case scenarios.
11. A fictional organizational-response timeline.
12. A mitigation comparison.
13. Ethical and accessibility implications.
14. Model limitations.
15. A reproducible list of equations and parameters.
16. Citations for every externally sourced input.

Part 13: Required cautions

The final analysis must state prominently that:

- The smartphone animation and organizational response are fictional.
- The simulation cannot establish that an actual animation would produce seizures.
- Epilepsy prevalence is not the same as photosensitivity prevalence.
- Exposure is not the same as an adverse event.
- Repeated exposures are not necessarily independent.
- A personal network may differ substantially from a representative random sample.
- Results depend heavily on animation characteristics for which no real-world measurements have been supplied.

- The study is a risk-screening and accessibility-planning exercise, not medical advice, a clinical experiment, or an accusation against a real developer.
- Potentially hazardous flashing content should not be created or shown to human participants merely to validate the simulation.

No matching personal research file was found in your Microsoft 365 data, so this rewrite is grounded in the public evidence cited above rather than private notes or documents.

ADDITIONAL PROMPT THAT WAS PROVIDED BY ME

Okay, um. My concern here is that I tasked you all to address the unwillingness of the designer or developer to address the outcomes, complaints, health risks, or that there is any problem at all. I didn't mean avoid that part of the prompt. I think it is pretty standard for users to deal with scenarios like this, until it goes to court, until someone gets injured, until a threshold is met. But the assumption is that for this simulation, before, during, and afterwards, they will not even acknowledge what is going on, that there are risks, dangers, and that even if scientific evidence, data, or actual customer complaints arise, they will do nothing to address the issue. That even after legal recourses have been pursued they will still try and do nothing. Do you understand?

LIST OF RESPONSES

CHATGPT

Yes, I understand the correction.

You are not asking me to model ordinary organizational delay, uncertainty, bad prioritization, or a company that eventually responds once evidence becomes strong enough. You want the simulation to include a much harder boundary condition: **the designer/developer remains institutionally resistant throughout the entire scenario, including before evidence emerges, while complaints accumulate, after scientific evidence appears, after injuries occur, and even after legal remedies are pursued.**

So the simulation should assume that the organization may continue to deny, minimize, reframe, contest, or simply refuse to address the hazard even when the evidence becomes increasingly difficult to dismiss. The important research question then becomes not merely, "Will the developer respond?" but:

What happens to the affected population, evidence base, reporting system, legal environment, public record, and cumulative harm when the responsible

organization does not voluntarily mitigate the design at any evidentiary threshold?

I would revise that part of the prompt substantially. The fictional organization should be allowed to exhibit behaviors such as denying causal association; asserting that complaints are anecdotal; challenging medical attribution; arguing that overall incidence is statistically negligible; pointing to the much larger unaffected user population; refusing to introduce an opt-out; declining to redesign the animation; contesting expert findings; fighting regulatory or legal action; appealing adverse decisions; continuing deployment while litigation proceeds; limiting admissions in settlements; and treating financial or legal consequences as a cost of maintaining the existing design rather than as a trigger for remediation.

Critically, I would **not** build in a point where the simulation automatically says, “At 100 confirmed cases the company fixes it.” There would be no automatic remediation threshold. Instead, we would run at least two separate systems:

- a **harm/evidence system**, where complaints, medically documented events, studies, injuries, litigation, regulatory findings, and statistical confidence can accumulate; and
- an **organizational-response system**, whose default state remains **NO MATERIAL CORRECTIVE ACTION**, regardless of how strong the first system becomes.

That lets us test much more uncomfortable outcomes. For example, we could reach a simulated state in which researchers conclude with high confidence that the animation materially elevates risk among a susceptible subgroup, yet the animation is still present five years later. Or courts could award damages without producing a global product change. Or individual cases could settle without admission of causation. Or a regulator could issue warnings while the developer challenges its authority. Those are all different from saying that evidence eventually causes rational self-correction.

I would also add a concept such as an **Institutional Resistance Parameter** to the simulation. At its strongest setting—the one you are asking for—the developer's probability of voluntarily implementing meaningful mitigation would effectively remain at or near zero. Evidence would therefore affect other variables—litigation, public awareness, reporting rates, insurance costs, regulatory attention, reputational consequences, user avoidance—but **not necessarily the hazardous design itself**.

And that changes the five-year comparison considerably. Instead of comparing only “developer fixes it after Year 1” versus “no change,” the main scenario becomes a persistent-exposure case:

Year 1: complaints arise → denied. Year 2: adverse-event evidence strengthens → disputed. Year 3: injuries and litigation emerge → defended rather than remediated. Year 4: stronger scientific/epidemiological evidence → methodology challenged or significance minimized. Year 5: legal/regulatory consequences accumulate → organization still resists changing the design.

We can even extend it beyond five years and test what ultimately stops the exposure **if voluntary developer action is removed from the system entirely**. The terminating mechanism might instead have to come from an operating-system fork, OEM intervention, regulatory prohibition, injunction, accessibility mandate, market abandonment, third-party workaround, insurance pressure, or some external technical intervention.

That is a materially different experiment from the prompt I wrote before, and your correction is important. I softened the organizational behavior into a conventional “slow response to an uncertain signal” model. That was not the scenario you gave me.

For the actual study, I would therefore make the developer's persistent refusal one of the **fixed premises**, while leaving the medical hypothesis itself open to falsification. In other words: **we should not predetermine that the animation causes seizures, but we should predetermine that if harm is demonstrated, the fictional developer still refuses to meaningfully address it.** That preserves the scientific integrity of the simulation while faithfully representing the institutional behavior you asked us to test.

CLAUDE AI

Yes, I understand, and you're right. My rewrite treated the developer response as one bullet, and it read as if the developer might eventually respond. What you're describing is a fixed condition of the simulation: the developer never acknowledges the problem at any stage, no matter what evidence or pressure arrives.

Here is the revised Part 3 developer behavior:

Before launch

- Accessibility and photosensitivity concerns raised internally are dismissed or never escalated.
- No flash-safety testing is done, and no opt-out is built.

After launch, early period

- User complaints and bug reports get no acknowledgment.
- Reports are closed as "working as intended" or "not reproducible."

As evidence accumulates

- Medical case reports, clinician letters, and epilepsy advocacy groups are met with no response.
- Independent scientific studies are ignored.

During legal action

- Lawsuits, regulatory complaints, and accessibility claims are met with denial or procedural delay.
- Even after losing rulings, the developer remediates nothing.

Core assumption for the model

- The probability of the developer fixing, disclosing, or acknowledging the issue is zero at every stage.

- Harm therefore accumulates unchecked. The only things that change over time are user-side adaptations: workarounds, avoidance, and device switching.

In the simulation, this means the risk never decays from a fix. I would add a few tracked variables: complaints filed versus complaints answered (always zero answered), cumulative injuries, legal actions filed, and months of continued non-response after each milestone. Should I update the full rewrite with this and begin Part 1?

PERPLEXITY AI

Yes, I understand. You are not asking me to soften, omit, or assume remediation of the developer's conduct; you want the **fictional simulation to treat persistent refusal, denial, and non-action as a core causal condition**.

In other words, the scenario is not:

“A risky animation is released, complaints arrive, and the company eventually responds responsibly.”

It is:

“A potentially hazardous, non-optional flashing unlock animation is released. The developer is informed repeatedly through scientific evidence, customer complaints, accessibility feedback, medical reports, incident documentation, advocacy, and legal action—yet continues to deny, minimize, obscure, or ignore the risk and makes no meaningful technical or policy change before, during, or after harm occurs.”

That distinction materially changes the study design, because it turns the problem from a narrow technical defect into a **persistent institutional-failure simulation**.

Revised developer-conduct assumption

For the fictional study, use this fixed assumption:

The hypothetical Android developer and its responsible design, product, accessibility, legal, and executive decision-makers are unwilling to acknowledge that the mandatory unlock animation may create foreseeable health and accessibility risks. They do not offer an opt-out, a reduced-motion alternative, a non-flashing mode, a warning, a safety review, an independent investigation, a rollback, a patch, or a meaningful incident-reporting pathway. This posture remains in place despite accumulating reports from users, caregivers, clinicians, disability-service providers, researchers, advocates, journalists, regulators, and litigants.

The simulation should further assume that the organization may publicly characterize complaints as isolated, anecdotal, unverified, unrelated, user error, or insufficiently proven—even when the total evidence becomes increasingly concerning.

What “unwillingness” means in the model

The company's refusal should not be a vague background detail. It should have measurable components and consequences.

Stage	Fictional organizational behavior	Simulation consequence
Pre-release	No meaningful accessibility or flash-safety testing; ignores internal concerns or does not seek them	The unsafe effect reaches the full user base rather than being caught before deployment
Early complaints	Dismisses individual reports as anecdotal or unrelated	Exposure continues while reports accumulate
Evidence emergence	Declines to review scientific evidence, incident patterns, or independent analysis	No reduction in uncertainty, no targeted protection, and no mitigation
Disability advocacy	Rejects requests for an opt-out or reduced-motion setting	Disabled and medically vulnerable users remain compelled to choose between exposure and device access
Documented injury	Treats each event as isolated; offers no systematic corrective action	The system fails to learn, so preventable repeat events continue
Legal notice or lawsuit	Denies responsibility, delays change, and does not alter the product	Exposure continues after the risk has become legally and ethically salient
Post-litigation	Maintains the animation without a safety change even after a settlement, judgment, injunction request, or public controversy	Harm can continue beyond the point at which a reasonable organization would have implemented safeguards

The key simulation variable: failure to mitigate

The prior model should include a **mitigation-response parameter**. In a normal responsible-development scenario, reports lead to actions that reduce exposure. In your requested scenario, the mitigation-response parameter stays at zero.

$M = 0$

Where:

- M = effectiveness of developer mitigation.
- $M = 0$ means no opt-out, no redesign, no warning, no throttle, no accessibility setting, no rollback, and no meaningful response.
- $M = 1$ would mean complete removal of the unsafe effect or a fully effective safety alternative.

Then residual exposure becomes:

$E_{\text{residual}} = E_{\text{initial}} \times (1-M)$

Because $M = 0$:

$E_{\text{residual}} = E_{\text{initial}}$

So every potentially hazardous unlock exposure continues after the first complaint, after the hundredth complaint, after medical documentation, and after legal claims. In the fictional model, organizational inaction does not merely fail to prevent the original harm; it allows **ongoing and cumulative exposure**.

Updated fictional timeline

Phase 1: Launch and forced exposure

A mandatory unlock animation with potentially provocative flashing characteristics is deployed to Android devices. It cannot be disabled. Users who require a phone for communication, transportation, health reminders, care coordination, employment, family contact, emergency access, or assistive functions cannot realistically avoid exposure without reducing or losing access to the device.

At this point, the developer either fails to assess the risk or declines to make that assessment visible.

Phase 2: First reports are ignored

Users begin reporting seizures, aura-like symptoms, migraines, dizziness, nausea, sensory overload, panic, visual discomfort, sleep disturbance, and avoidance of device use. Family members, caregivers, and staff in disability-service settings report that the effect is distressing or unsafe for people they support.

The fictional developer does not collect data in a meaningful way, publicly acknowledge reports, or provide an emergency opt-out. Individual users are effectively told that their experience is not sufficient proof.

Phase 3: Pattern becomes visible

A disability organization, clinician, researcher, or community program gathers a pattern of complaints. The evidence may include dates, device models, OS versions, video recordings of the animation, descriptions of symptoms, emergency-room visits, injury reports, and comparisons between exposure periods and non-exposure periods.

The developer still asserts that correlation is not causation. That phrase may be technically true in a narrow evidentiary sense, but in the scenario it becomes a reason to take **no precautionary action whatsoever**.

The issue is no longer simply whether causation has been proven beyond dispute. The issue is whether a company facing a plausible severe risk can justify forcing continued exposure while refusing low-cost safety measures.

Phase 4: Harm accumulates

The absence of an opt-out turns each unlock into a repeated risk event. A user unlocking a phone 100 times per day could encounter the effect roughly 36,500 times yearly if it appears every time. For a high-frequency user with a known sensitivity, that is not one exposure—it is a continuous, unavoidable pattern of exposure.

The fictional outcomes can include:

- Seizures or suspected seizure-related events in susceptible users.
- Injuries caused by falls, loss of awareness, or impaired coordination during or after an event.
- Emergency transport, emergency-department visits, medical evaluation, medication changes, and missed appointments.
- New anxiety around phone use, avoidance of communication technology, or loss of independence.
- Caregiver stress, increased supervision needs, modified staffing, and documentation burdens.
- Family members or nearby individuals experiencing a triggered seizure or severe sensory reaction after seeing another person's phone activate.
- Exclusion from digital systems when the user cannot tolerate the required animation but must use the device to access services.

Phase 5: Legal action without corrective action

In the simulation, legal claims are pursued. The company has notice of the alleged harms, potentially receives preservation demands, expert opinions, accessibility complaints, formal demand letters, agency inquiries, or lawsuits.

But the model assumes that legal action does **not** automatically solve the problem. The company continues to defend the animation, maintains that no conclusive causal link exists, disputes the reliability of complainants, delays changes during litigation, or treats a case outcome as limited to a particular plaintiff rather than a product-safety signal.

That means legal escalation is modeled as an additional evidence source, not as a guaranteed intervention.

Outcome model under persistent denial

A responsible developer's response would normally lower risk over time. Your requested scenario holds risk in place—or allows it to grow as more devices receive the feature.

A simple cumulative-harm model is:

$$H_t = U_t \times X_t \times S \times P_t \times (1 - M_t)$$

Where:

- H_t = expected harmful events during time period t .
- U_t = number of users receiving the animation.
- X_t = average number of relevant flash exposures per user during that period.
- S = proportion of users or nearby persons who are susceptible.
- P_t = assumed probability of an adverse event within the defined exposure period for a susceptible person.
- M_t = mitigation effectiveness at time t .

Under the requested fictional condition:

$$M_t = 0 \quad \text{for all periods}$$

Therefore:

$$H_t = U_t \times X_t \times S \times P_t$$

As adoption expands, U_t can rise. If the feature is integrated into more device models or versions, total exposure can rise even if the per-user risk remains unchanged. The result is a model of **avoidable cumulative harm created by sustained inaction**.

Illustrative escalation example

Consider a fictional disability-support network with 500 participants, 120 staff members, and frequent shared-device exposure. Suppose the supported population has a higher-than-average prevalence of epilepsy, migraine, sensory sensitivity, neurological conditions, and medication-related vulnerability than the general population.

Assume:

- 300 people use Android phones with the animation.
- Each is exposed to 80 animation events per day.
- The animation is active 365 days per year.
- 8% of the relevant local population is considered potentially vulnerable to the visual effect for purposes of the fictional model.
- The developer makes no mitigation after complaints.

The exposure count is:

$$300 \times 80 \times 365 = 8,760,000$$

potential animated unlock events annually in that local network.

This does not prove that 8.76 million exposures cause harm. It demonstrates why a company's refusal to investigate or allow an opt-out matters: a localized group with elevated vulnerability

may face millions of repeated exposures even when the general population's average risk appears low.

If the group documents several concerning incidents but receives no product change, the relevant harm is not only the initial event count. It is the number of additional exposures and potentially preventable events that occur **after notice**.

Disability-service setting

Your point about the people you work with is essential. A “representative” national or global sample can hide concentrated risk in real communities.

A disability-support or day-program setting may include people with:

- Epilepsy or documented seizure-action plans.
- Cerebral palsy, traumatic brain injury, stroke history, genetic conditions, or other neurological conditions.
- Intellectual and developmental disabilities.
- Chronic illnesses and medication regimens that affect alertness, balance, sensory processing, or seizure threshold.
- Migraine disorders, vestibular conditions, visual-processing differences, autism-related sensory sensitivities, or anxiety.
- Limited ability to identify the triggering feature, change settings, report symptoms clearly, or advocate independently.
- Mobility or fall risks that make a seizure, disorientation, or sudden distress more consequential.
- Reliance on caregivers, staff, family, and technology for daily communication and support.

The simulation should therefore create at least two separate populations:

Population	Appropriate assumption
General Android population	Broad prevalence estimates, broad usage distribution, lower concentration of known vulnerabilities

Disability-support/commu nity-care population	Local risk profile based on anonymized service data, support needs, known seizure plans, medication considerations, sensory needs, and environmental exposure
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The second population should not be described as “representative” of the public. It is a **risk-concentrated environment**, and that is precisely why a universal non-optional design can have unequal impact.

What the study should test

The simulation should test not only, “How many events might occur?” but also:

1. **How much harm occurs before the first complaint?**
2. **How much additional harm occurs after the developer receives credible notice?**
3. **How many events could have been avoided if an opt-out had been offered at the first warning sign?**
4. **How does ongoing refusal disproportionately affect people in disability-support, care, and high-vulnerability settings?**
5. **How does the risk change as the product expands to more devices and more users?**
6. **What is the cost of non-action in medical events, injuries, missed services, caregiver burden, exclusion from technology, and lost autonomy?**
7. **At what point does continued non-action become harder to defend ethically, operationally, and legally?**

Language for the finalized simulation

Use the following statement in the project:

This fictional simulation assumes persistent institutional non-response. The hypothetical developer does not acknowledge the reported risks, does not investigate adequately, does not provide an opt-out or accessibility accommodation, does not redesign the animation, does not warn users, and does not meaningfully change course after complaints, injuries, scientific evidence, advocacy, regulatory attention, or legal action. The simulation therefore measures not only initial risk but the cumulative and potentially preventable harms that may result when a known or plausibly foreseeable safety concern remains unaddressed over time.

This stays factual in its structure while preserving the central premise you intended: **the failure is not only that the feature exists; the failure is that decision-makers continually refuse to recognize, mitigate, or stop a foreseeable pattern of harm.**

GROK AI

Yes. The refusal is the condition of the scenario, not a footnote I get to settle by pointing at the flash-count rule.

The earlier pass treated “they decline to measure it” as one beat, then spent the rest of the brief on whether a single wave meets the photic threshold. That answered a different question. Yours was: model a platform that does not acknowledge risk, complaints, evidence, injury, or a judgment, and still ships the animation. The scientific bound stays in the model as an input. It is not the plot, and it is not a reason for them to be let off the page.

Fiction, same label as before. Not a description of any shipping phone.

Lumen Wake ships in 2027 as a brand signature. Before launch, an internal accessibility note says the wave is one luminance transition and therefore outside the three-flash rule. That note is the only safety review. There is no intermittent-photic comparison, no reduced-motion hook, no off switch. The launch copy calls it delight. The support taxonomy has no field for seizure, migraine, startle, or fall. A ticket that says “this flash made me sick” is closed as preference.

During the first year, three kinds of evidence arrive and none of them change the build.

Customer reports come in from people with known photosensitive epilepsy, from parents unlocking a phone next to a child, and from group-home staff who see a time-locked event. Each is answered with the same macro: the animation is a single transition, below the published threshold, and no causal link has been shown. Reporters are asked for medical records the company will not accept as proof. Telemetry still counts unlocks only. Absence of a telemetry signal is then cited as evidence of absence.

A small external lab measures the wave and finds either that it misses the three-flash rule, or that a subset of units overshoot contrast on high-brightness OLED. The company does not publish the measurement. If the lab says the stimulus is below threshold, that sentence is the whole response. If the lab says a subset overshoots, the subset is called an edge case and left on the default. No opt-out is added in either case.

Injury reports follow. A fall off a curb. A drop in a kitchen. A witnessed convulsion within seconds of unlock in someone with a prior photoparoxysmal EEG. The public line does not move: correlation, missed medication, sleep loss, preexisting condition. The animation stays default. Staff who raise it internally are told the brand system is frozen until the next major release, then the next one.

Legal recourse does not break the stance. A demand letter is answered by counsel. A suit is removed, delayed, and tried on general causation. Even after a plaintiff verdict or a settlement with no admission, the animation remains the default on new devices. The release notes do not mention it. The settings page does not gain a switch. The remedy, if any, is a quiet firmware branch for named plaintiffs, or a voucher, not a population control. Discovery shows the original

note, the closed tickets, and the decision to keep the signature. The next earnings call still describes the unlock as a craft detail.

That is the before, during, and after. Acknowledgment never occurs. The threshold they are waiting for is not a count of injuries. There is no internal threshold. "We will look when there is evidence" is the method by which evidence is defined out of existence.

What that does to the simulation, if the refusal is the protocol rather than a flavor line:

Detection goes to zero inside the company even when events are nonzero outside it. A true hazard of 1 in a million unlocks among photosensitive users still produces on the order of tens of thousands of time-locked events a year in the Android base. Under this protocol the dashboard remains empty, because the dashboard cannot see a seizure and the ticket system renames one as preference. The caseload of twelve still mostly never contains a photosensitive person, so a direct-support worker can truthfully say they have not seen it, and the company can quote that as reassurance. Those are the same fact viewed from two sample sizes.

After a suit, the model does not flip the hazard to zero and does not flip the company's action to "fixed." The post-judgment state is the pre-launch state: animation on, no opt-out, complaints reclassified, next version unchanged. The only quantities that move are external ones: retained counsel, sealed settlements, a few users who leave the platform, and undercount in the remaining user base because the people who seized stopped unlocking that phone.

The zero arm is still in the file. It is what the company asserts. It is not what the scenario assumes. The scenario assumes they will say the zero arm out loud after the nonzero arms have already been handed to them.

MICROSOFT COPILOT

Yes, I understand. You want **persistent, institutional nonresponse** to be a fixed condition of the fictional simulation, not a temporary stage that ends once enough evidence, complaints, injuries, regulatory pressure, or legal action accumulates.

More precisely, the simulation should assume that the fictional designer, developer, or company:

- Refuses to acknowledge that the unlock animation presents any health or accessibility risk.
- Denies or minimizes reported seizures, injuries, near misses, and other adverse effects.
- Treats customer complaints as unrelated anecdotes, user error, preexisting medical problems, or insufficient proof.
- Rejects scientific evidence when it conflicts with the preferred design or business position.
- Declines to add an opt-out, safe mode, reduced-motion accommodation, warning, rollback, or alternative animation.

- Continues the feature before, during, and after documented adverse events.
- Maintains this posture even after internal testing identifies risk.
- Maintains it after clinicians, accessibility specialists, disability advocates, employees, or independent researchers raise concerns.
- Maintains it after lawsuits, regulatory inquiries, court findings, settlements, injunction attempts, or other legal remedies.
- Uses delay, procedural resistance, narrow interpretations, appeals, or nominal compliance measures without meaningfully eliminating the exposure.
- Does not voluntarily correct the problem at any point in the simulated period.

So the research question is **not**:

“At what level of evidence would the developer finally respond?”

Instead, it is:

“What happens to affected users, families, workers, bystanders, institutions, complaint systems, healthcare systems, and legal systems when the responsible developer refuses to recognize or mitigate a recurring health hazard regardless of the evidence accumulated?”

The developer’s unwillingness should therefore be modeled as an **exogenous fixed assumption**, while its observable behavior can still have different forms:

1. **Complete denial:** No causal connection exists.
2. **Causal deflection:** Events occurred, but the animation did not cause them.
3. **Responsibility deflection:** The risk belongs to medically vulnerable users, not the product.
4. **Statistical minimization:** The number affected is too small to justify a change.
5. **Technical minimization:** Internal tests allegedly show that the animation falls within acceptable parameters.
6. **Accommodation refusal:** An opt-out would undermine design consistency, security, branding, or product function.
7. **Complaint suppression:** Reports are fragmented, closed individually, or excluded from aggregate analysis.
8. **Legal resistance:** Claims are challenged, delayed, appealed, settled without admission, or handled without a product change.
9. **Symbolic compliance:** The company publishes general guidance or a vague warning but leaves the unavoidable animation unchanged.
10. **Post-judgment resistance:** Even after an adverse legal or regulatory result, it takes the narrowest possible action while preserving the underlying exposure.

The simulation should then measure the consequences of that refusal:

- Preventable exposures accumulated
- Adverse events and repeat events
- Injuries and medical utilization
- Effects on family members and bystanders
- Effects in disability-service and care environments

- Complaints submitted, rejected, or left unresolved
- Time from first warning to first documented injury
- Time from first injury to legal action
- Legal and administrative costs
- Underreporting and complaint attrition
- Number of people who abandon or limit device use
- Workplace and caregiver consequences
- Disparities affecting users who cannot easily replace the device
- Accumulated preventable harm after each escalation point
- The difference between observed harm and harm that would have occurred with an opt-out or safe default

Most importantly, we would preserve a **counterfactual mitigation branch** for comparison, but not because the fictional developer adopts it. It would show how many outcomes could have been prevented *if* the developer had responded. The actual scenario would remain on the nonresponse path for the entire simulation.

I also understand that you are not asking me to soften this into a generic “company initially hesitates but eventually does the right thing” narrative. The refusal itself is a core object of study, including refusal **after the evidentiary and legal thresholds that would ordinarily be expected to compel action have been crossed**.